

ARUL K A

AI ENGINEER

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SUMMARY

AI Engineer with hands-on experience in developing intelligent applications using Machine Learning, Deep Learning, Large Language Models, Retrieval-Augmented Generation, and Conversational AI technologies. Skilled in designing, fine-tuning, and deploying AI solutions across cloud and edge environments, with expertise in prompt engineering, information retrieval systems, computer vision, speech technologies, and AI model optimization. Passionate about transforming complex challenges into practical AI-driven solutions while embracing the philosophy of Engineering Intelligence Through Intelligence.

2019 – 2023

MAHENDRA ENGINEERING COLLEGE

Bachelor of Engineering in Mechanical and Automation.

CERTIFICATES

2024

DATA SCIENCE

KG Micro college.

2024

IBM – DATA SCIENCE

Coursera

SKILLS

- Prompt Engineering
- Git & Version Control
- Voice AI & Conversational AI Systems
- AI Model Training & Fine-Tuning
- Python
- Machine Learning
- Deep Learning
- Data Preprocessing & Feature Engineering
- Large Language Models (LLMs)
- Natural Language Processing (NLP)
- Retrieval-Augmented Generation(RAG)
- Database Querying & Vector Databases
- Offline AI Model Deployment
- Linux (Ubuntu)
- Web Scraping
- SQL
- Microsoft365, Excel
- Tableau, Microsoft Power BI
- Hyperparameter Optimization

Tools & Frameworks

- FastAPI
- PyTorch
- FAISS / ChromaDB
- CUDA
- Hugging Face
- TensorFlow
- OpenCV

Experience

- Developed and deployed AI-powered applications using Large Language Models, Retrieval-Augmented Generation (RAG), Conversational AI, Computer Vision, and Speech Technologies.
- Fine-tuned foundation models using domain-specific datasets and optimized AI solutions for cloud and edge deployment environments.
- Implemented prompt engineering, information retrieval systems, AI guardrails, and model optimization techniques for production-ready applications.
- Collaborated on the development of multimodal AI systems integrating Large Language Models, Computer Vision, and Speech Technologies.

AI Engineer - App Innovation Technologies, Coimbatore, Tamil Nadu

Jun 2025 – Present

Projects

Power connect

Developed an intelligent document retrieval and conversational AI platform by processing web-scraped and enterprise data sources. Built Retrieval-Augmented Generation (RAG) pipelines with hybrid retrieval mechanisms, applied prompt engineering techniques, implemented sentiment analysis and automated file generation workflows, and integrated API-based large language models to deliver context-aware responses.

Realityworks AI

Configured GPU-accelerated environments for offline deployment of speech and language models. Fine-tuned foundation models using domain-specific datasets, implemented speech enhancement techniques, integrated offline STT, TTS, and language model pipelines, and optimized inference workflows for local deployment scenarios.

Debris IQ (DDRP)

Worked on extracting and processing roadway-related data for machine learning applications across the United States. Performed feature engineering, managed structured data storage using SQL databases, generated training datasets, developed Random Forest models, applied computer vision techniques for image classification, and integrated weather intelligence data to enhance analytical capabilities.

Onassist AI

Contributed to the development of a multimodal AI assistant by integrating computer vision models for avatar generation, implementing voice-enabled interactions, and applying domain-specific guardrails. Enhanced overall user engagement through seamless vision, speech, and language model integration.

Jetson AI

Developed and optimized AI solutions for NVIDIA Jetson edge devices by configuring CUDA-enabled environments and deployment pipelines. Built offline voice assistant capabilities, fine-tuned AI models using domain-specific datasets, and optimized inference performance for resource-constrained edge computing environments.